

RAJESH MAYILSAMY

📍 Linköping, Sweden

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EDUCATION

Master of Science in Aeronautical Engineering

August 2025 - Present

Linköping University, Sweden

Coursework: Aerodynamics, Computational Fluid Dynamics, Aircraft Conceptual Design, Computational Aerodynamics, Finite Element Methods, Flight Mechanics, Gas Turbines

SKILLS AND INTERESTS

Interests	Computational Fluid Dynamics, Turbomachinery, Gas Turbines, Heat Transfer, Design and Analysis, Modelling and Simulation
Design Software	AutoCAD, CATIA V5, ANSA, Hopsan, ANSYS (Static Structural, CFX, Fluent), MATLAB and Simulink, Programming in C, C++, Python
Languages	English (full professional), Tamil (native), German (A1), Swedish (learning)

EXPERIENCE

Sree Murugan Garments

July 2022 - April 2025

Production Manager - Full-Time

Oversaw daily production operations, raw material sourcing, quality inspections and cross-department coordination to ensure efficient manufacturing and adherence to quality standards.

Vyadh Aerospace Pvt. Ltd.

September 2023 - November 2023

CFD Intern and Trainee (Part-Time)

Worked on meshing, solver setup, aerodynamic simulations and performance analysis during a 3-month CFD internship.

Simulation Lab Pvt. Ltd.

April 2022 - June 2022

CAE Engineer Intern

Contributed to aerodynamic product development through computational simulations and supported a patent application on wing flap efficiency enhancement.

PROJECTS

Large Eddy Simulation of Flow over a Periodic Hill

May 2026 - July 2026

- Ran LES in ANSYS Fluent on the university HPC cluster using Linux; extracted results and compared against DNS benchmark data, using MATLAB to process EnSight binary output.

External Aerodynamics: CFD Analysis of Cybertruck Aerodynamics

March 2026

- Simulated the external aerodynamic flow of a Tesla Cybertruck geometry in ANSYS Fluent; verified and validated the model against reference data.

Aircraft Conceptual Design - Project Course

March 2026 - June 2026

- Conceptualised a VTOL tilt-rotor delivery aircraft for a company named Bussgods, with a design mission range of around 160 km and a 20 kg payload; carried out range and endurance, specific excess power, CG sweep and payload-range analysis in MATLAB.

Design and Multi-perspective Investigation on Nature-Inspired

August 2021 - June 2022

Unmanned Amphibious Vehicle for Rescue Application

Bachelor Thesis

- Designed a rescue-oriented unmanned amphibious vehicle inspired by the devil ray fish, performing CFD and FEA analysis to evaluate aerodynamic, hydrodynamic, and structural performance.

RESEARCH PUBLICATION

Conceptual Design and Contour Optimization of Altitude Compensation Nozzles for SSTD Vehicle, published at the *AIAA Propulsion and Energy Forum* (2020), AIAA 2020-3922.